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WearWise: An Intelligent Knowledge-Based System for Sustainable Wardrobe Management

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Domain & Motivation



The fashion industry faces two major problems:

- People own many clothes but struggle to choose outfits
- Most garments are underused
- Textile waste is growing rapidly
- Daily outfit decisions create cognitive overload

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- Decision fatigue
- Low wardrobe utilization
- Unsustainable consumption habits

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Help users make better daily outfit decisions using the clothes they already own.

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Objectives & System Scope



Current fashion apps mainly:

- Recommend new products
- Are based on trends or sales
- Do not reason about context
- Do not explain decisions

- Is this outfit appropriate for today's weather?
- Is it suitable for the occasion?
- Does it respect formality and season rules?

Design an intelligent knowledge-based system that reasons like a human stylist in order to maximize utility of existing wardrobe

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Knowledge-Based System Foundations

Knowledge Base

Facts and rules about fashion, climate, and social contexts

Facts Base

Contains the current facts describing the garments and the user’s situation.

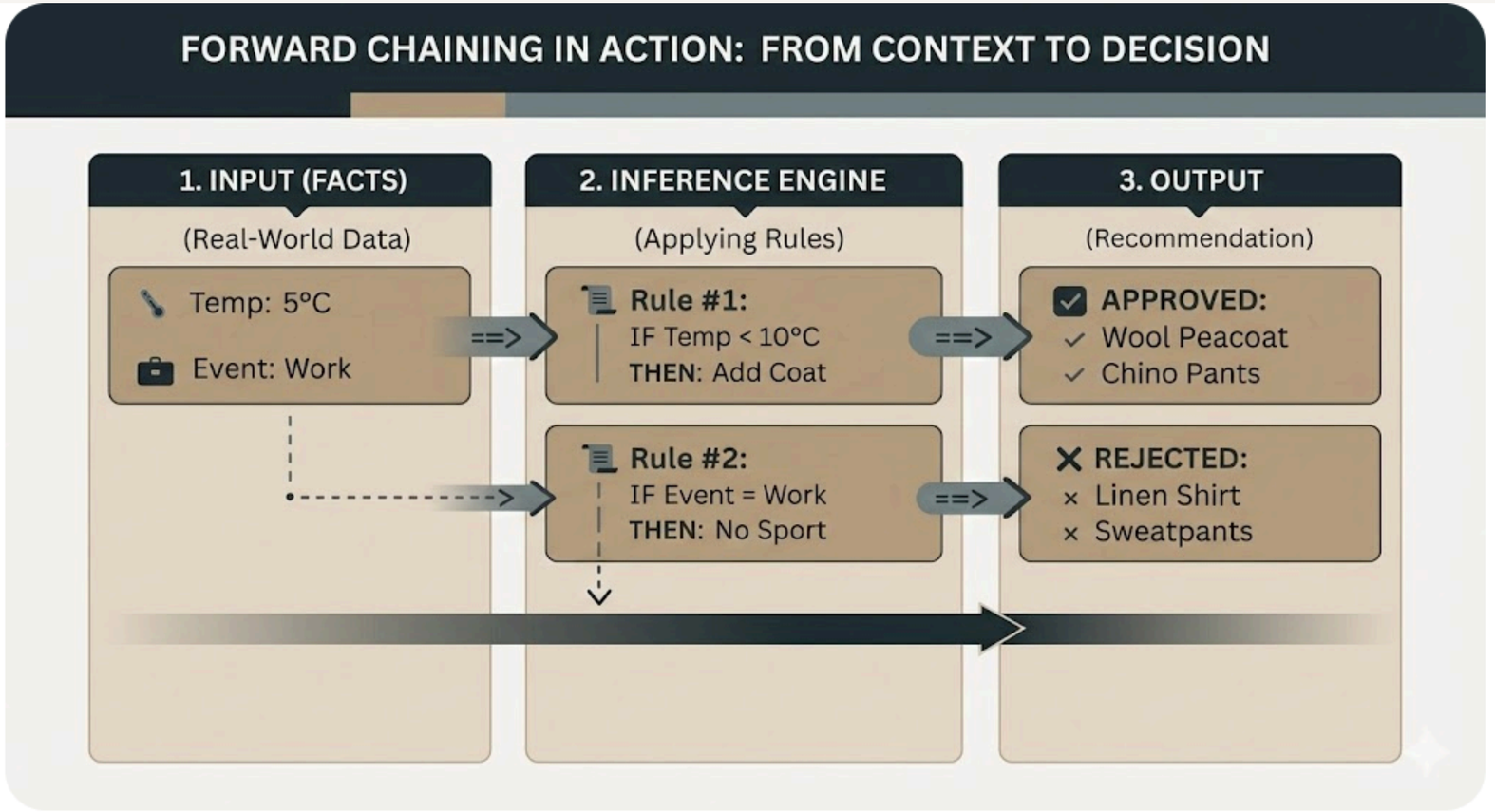
Inference Engine

Processes knowledge to generate logical outfit recomendations

Key Insight

We need logic, constraints, and explanations, not predictions.

Knowledge-Based System Foundations



Key Insight

Logic filters 50+ items down to 3 valid ones

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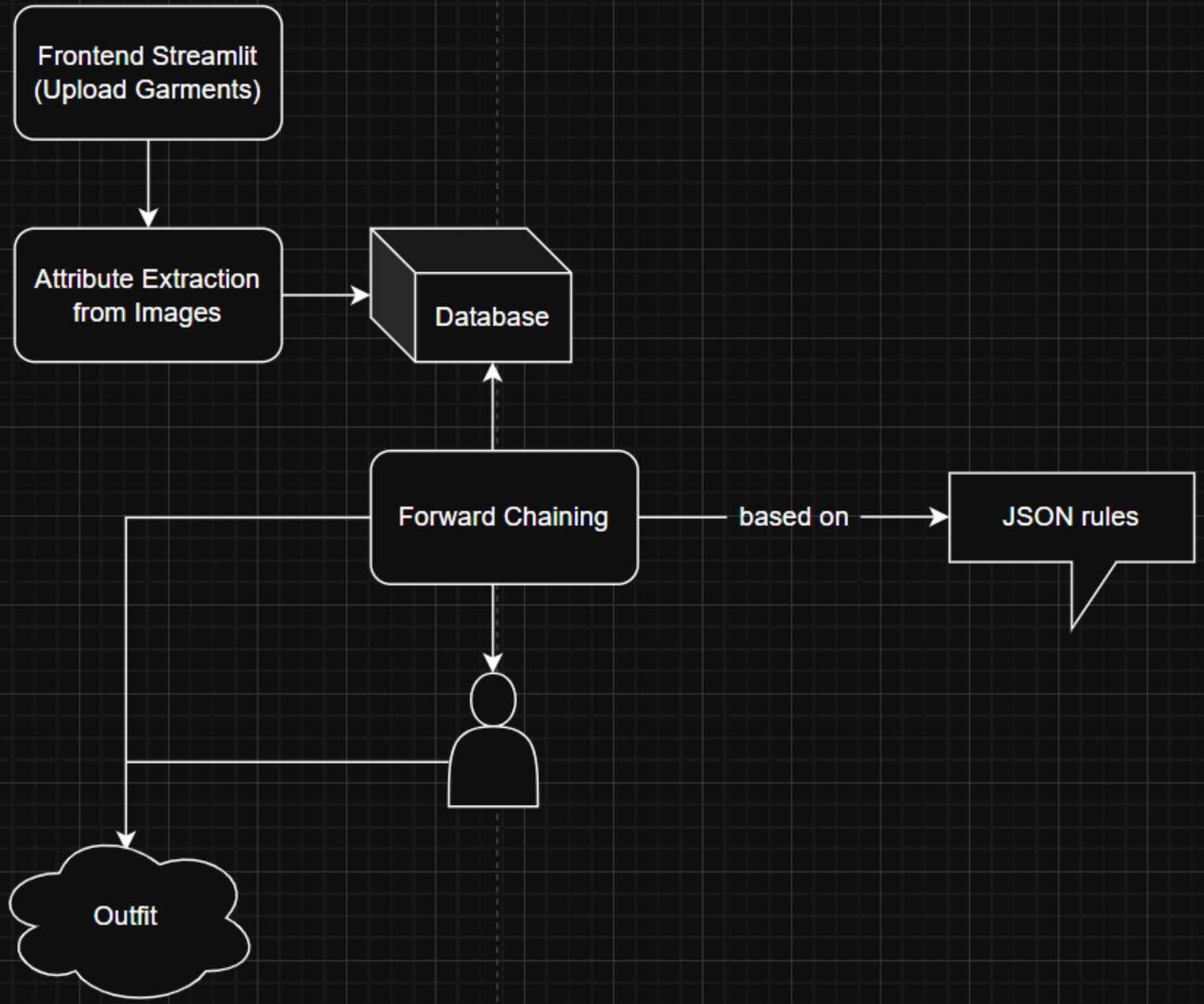
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Global Architecture Overview



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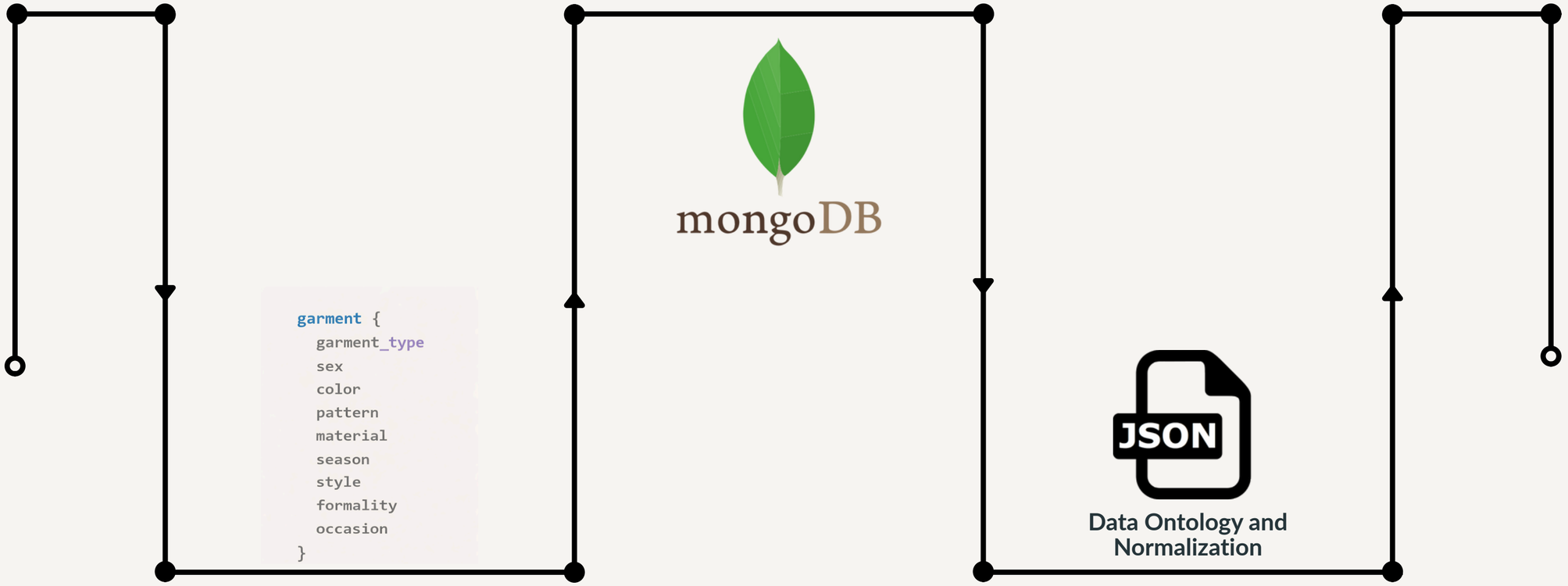
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Data Engineering and Persistance Layer



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Automatic Knowledge Acquisition

Extracted attributes are mapped to a controlled vocabulary to ensure consistent and reliable values for reasoning.

Image Input → AI extraction → Normalization → Database Storage

Users upload images of garments from their personal wardrobe to be analyzed by the system

GPT analyzes each image and extracts attributes such as color, material, season, and style

The normalized garment data is stored as structured knowledge, forming the knowledge base used by the inference engine

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Inference Engine and Rule Structure

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01

Forward Chaining

- Python-based algorithm
- Builds outfit solution from facts
- Rules can be easily modified



02

Rule Application

- User context forms initial facts
- Rules infer derived facts
- Rules chain together

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Recomendation and Scoring Process

01



Logical filtering

Initial filtering based on strict rules with attributes tagged with “forbidden” (ex: excluding shorts for cold weather).

02



Heuristic Scoring

Valid candidates are ranked using a scoring system that considers soft rules (ex: color harmony, formality) and hard rules.

03



Explainable Decisions

Every recommendation comes with a trace of activated rules and applied heuristics, ensuring transparency.

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Conclusion + Real-Time example

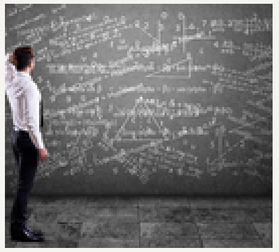
01



Limitations:

The system depends on image extraction quality and a handcrafted rule base, which limits coverage of styles and personal preferences.

02



Conclusions:

WearWise demonstrates that a knowledge-based system can generate explainable and context-aware outfit recommendations.

03



Future Improvements

Future work includes expanding the rules, improving extraction accuracy, and adding user preference and real-time data integration.

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